

Case study 10:-

Restaurant POS (Point of Sale) System:- A restaurant wants to automate billing. Accept quantities and prices of Pizza, Burger, and Cold Drink, calculate Subtotal, GST (18%), and Grand Total. Display a professional invoice.

Algorithm – Restaurant POS (Point of Sale) System

1. Start.
2. Declare variables: pizzaQty, burgerQty, drinkQty, pizzaPrice, burgerPrice, drinkPrice, subtotal, gst, grandTotal.
3. Input the quantity and price of Pizza.
4. Input the quantity and price of Burger.
5. Input the quantity and price of Cold Drink.
6. Calculate the item totals:
 - $\text{pizzaTotal} = \text{pizzaQty} \times \text{pizzaPrice}$
 - $\text{burgerTotal} = \text{burgerQty} \times \text{burgerPrice}$
 - $\text{drinkTotal} = \text{drinkQty} \times \text{drinkPrice}$
7. Calculate the Subtotal:
 - $\text{subtotal} = \text{pizzaTotal} + \text{burgerTotal} + \text{drinkTotal}$
8. Calculate GST (18%):
 - $\text{gst} = \text{subtotal} \times 18 / 100$
9. Calculate Grand Total:
 - $\text{grandTotal} = \text{subtotal} + \text{gst}$
10. Display the invoice with item totals, Subtotal, GST, and Grand Total.
11. Stop.

Pseudocode:-

BEGIN

DECLARE pizzaQty, burgerQty, drinkQty AS INTEGER

DECLARE pizzaPrice, burgerPrice, drinkPrice AS FLOAT

DECLARE pizzaTotal, burgerTotal, drinkTotal AS FLOAT

DECLARE subtotal, gst, grandTotal AS FLOAT

INPUT pizzaQty, pizzaPrice

INPUT burgerQty, burgerPrice

INPUT drinkQty, drinkPrice

pizzaTotal = pizzaQty * pizzaPrice

burgerTotal = burgerQty * burgerPrice

drinkTotal = drinkQty * drinkPrice

subtotal = pizzaTotal + burgerTotal + drinkTotal

gst = subtotal * 18 / 100

grandTotal = subtotal + gst

PRINT "----- Restaurant Invoice -----"

PRINT "Pizza Total = ", pizzaTotal

PRINT "Burger Total = ", burgerTotal

PRINT "Cold Drink Total = ", drinkTotal

PRINT "Subtotal = ", subtotal

PRINT "GST (18%) = ", gst

PRINT "Grand Total = ", grandTotal

END

Test cases:- input/output combination.

```
PS C:\Users\khush\Desktop\PROJECT 1> gcc CASE_study_10.c -o CASE_study_10 ; if ($?) { .\CASE_study_10 }
Enter Pizza Quantity: 2
Enter Burger Quantity: 1
Enter Cold Drink Quantity: 2

===== RESTAURANT INVOICE =====
Pizza      : 2 x 250 = 500.00
Burger     : 1 x 120 = 120.00
Cold Drink : 2 x 60  = 120.00
-----
Subtotal   : 740.00
GST (18%)  : 133.20
Grand Total: 873.20
=====
PS C:\Users\khush\Desktop\PROJECT 1> 
```

```
PS C:\Users\khush\Desktop\PROJECT 1> gcc CASE_study_10.c -o CASE_study_10 ; if ($?) { .\CASE_study_10 }
Enter Pizza Quantity: 4
Enter Burger Quantity: 2
Enter Cold Drink Quantity: 3

===== RESTAURANT INVOICE =====
Pizza      : 4 x 250 = 1000.00
Burger     : 2 x 120 = 240.00
Cold Drink : 3 x 60  = 180.00
-----
Subtotal   : 1420.00
GST (18%)  : 255.60
Grand Total: 1675.60
=====
PS C:\Users\khush\Desktop\PROJECT 1> 
```

```
PS C:\Users\khush\Desktop\PROJECT 1> gcc CASE_study_10.c -o CASE_study_10 ; if ($?) { .\CASE_study_10 }
Enter Pizza Quantity: 3
Enter Burger Quantity: 1
Enter Cold Drink Quantity: 3

===== RESTAURANT INVOICE =====
Pizza      : 3 x 250 = 750.00
Burger     : 1 x 120 = 120.00
Cold Drink : 3 x 60  = 180.00
-----
Subtotal   : 1050.00
GST (18%)  : 189.00
Grand Total: 1239.00
=====
PS C:\Users\khush\Desktop\PROJECT 1> 
```